



CANDIDATE – PLEASE NOTE!

PRINT your name on the line below and return this booklet with your answer sheet. Failure to do so may result in disqualification.

FORM TP 02025097

TEST CODE 01234010

MAY/JUNE 2025

**CARIBBEAN EXAMINATIONS COUNCIL
CARIBBEAN SECONDARY EDUCATION CERTIFICATE®
EXAMINATION**

MATHEMATICS

Paper 01 – General Proficiency

1 hour 30 minutes

06 JUNE 2025 (a.m.)

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

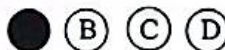
1. This test consists of 60 items. You will have 1 hour and 30 minutes to answer them.
2. In addition to this test booklet, you should have an answer sheet.
3. A list of formulae is provided on page 2 of this booklet.
4. Each item in this test has four suggested answers lettered (A), (B), (C), (D). Read each item you are about to answer and decide which choice is best.
5. On your answer sheet, find the number which corresponds to your item and shade the space having the same letter as the answer you have chosen. Look at the sample item below.

Sample Item

$$2a + 6a =$$

- (A) $8a$
- (B) $8a^2$
- (C) $12a$
- (D) $12a^2$

Sample Answer



The best answer to this item is “ $8a$ ”, so (A) has been shaded.

6. If you want to change your answer, erase it completely before you fill in your new choice.
7. When you are told to begin, turn the page and work as quickly and as carefully as you can. If you cannot answer an item, go on to the next one. You may return to that item later.
8. You may do any rough work in this booklet.
9. Calculators and mathematical tables are NOT allowed for this paper.
10. Diagrams in this booklet are NOT drawn to scale, unless otherwise stated.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

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1. $(-3)^2 + (-2)^2$ is equal to

(A) -13
(B) -10
(C) 13
(D) 25

2. What percentage of 40 is 50?

(A) 20%
(B) 25%
(C) 80%
(D) 125%

3. What number when added to $1\frac{1}{3}$ gives 2?

(A) $\frac{1}{3}$
(B) $\frac{2}{3}$
(C) 1
(D) 3

4. If $235 \times 48.7 = 11\,444.5$, then

$$0.235 \times 48.7 =$$

(A) 11.4445
(B) 114.445
(C) 1 144.45
(D) 11 444.4

Item 5 refers to the following sequence.

1, 6, 13, 22, 33, _____.

5. If the pattern in the sequence continues, then the next term would be

(A) 44
(B) 45
(C) 46
(D) 52

6. Dan sold a certain number of concert tickets in 3 days. The ratio of the number of tickets sold over the 3 days is 3:7:5 respectively. If he sold 8 MORE tickets on Day 2 than on Day 3, the number of tickets he sold ALTOGETHER is

(A) 40
(B) 60
(C) 72
(D) 120

7. Which of the following sets is defined by $\{x \in \mathbb{Z} : -2 \leq x \leq 4\}$?

(A) $\{1, 2, 3, 4\}$
(B) $\{0, 1, 2, 3, 4\}$
(C) $\{-1, 0, 1, 2, 3\}$
(D) $\{-2, -1, 0, 1, 2, 3, 4\}$

8. Which of the following is an example of two-digit positive integers that are divisible by 6?

(A) A finite set
(B) An empty set
(C) An infinite set
(D) An improper set

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9. If $n(U) = 25$, $n(A) = 14$, $n(B) = 15$ and $n(A \cap B) = 6$, then $n(A \cup B)$ is

(A) 2
(B) 8
(C) 9
(D) 23

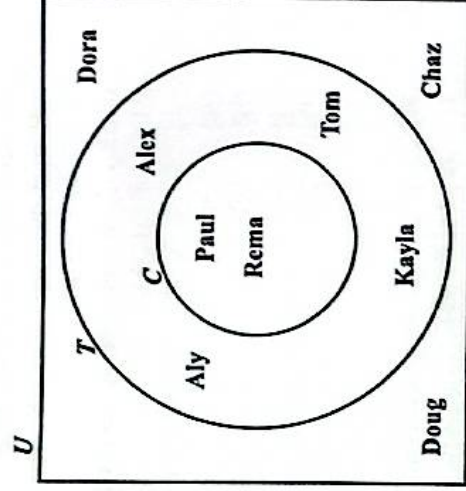
Item 11 refers to the following information on the description of 3 sets.

$Q = \{\text{odd numbers}\}$
 $R = \{\text{even numbers}\}$
 $P = \{\text{prime numbers}\}$

11. Which of the following sets is empty?

(A) $P \cap R$
 (B) $P \cup Q$
 (C) $P \cap Q$
 (D) $Q \cap R$

Item 10 refers to the following Venn diagram which shows the universal set (U), and two sets, T and C , that represent the 9 students in a class, some of whom play tennis (T) and chess (C).



10. How many students play tennis but NOT chess?

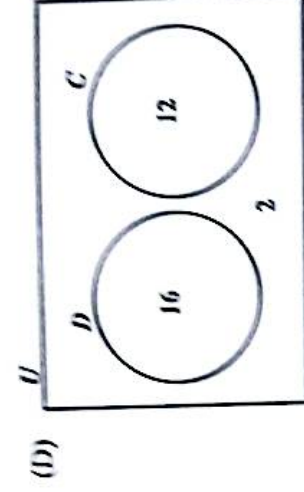
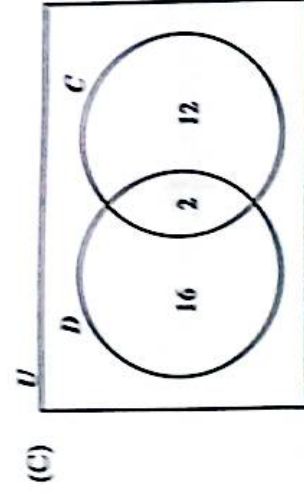
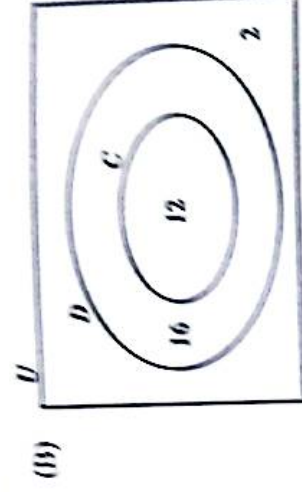
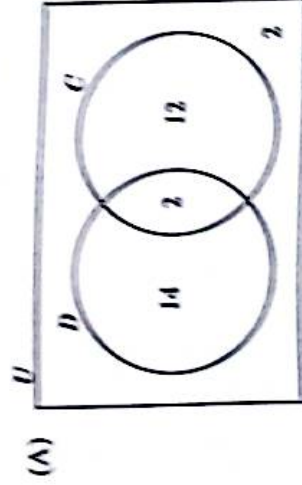
(A) 2
(B) 3
(C) 4
(D) 6



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12. The 30 students in Teacher May's class have either a dog or a cat or none of the two. Sixteen students have a dog (D), 12 students have a cat (C) and the remainder have neither a cat nor a dog. Which of the following Venn diagrams correctly represents this information?



13. At a bank, EC\$2.60 is equivalent to US\$1.00. For every US\$1.00 exchanged, EC\$0.10 is deducted as an exchange fee. How much EC dollars will Leon receive if he exchanges US\$1 000.00?

- (A) \$ 900.90
(B) \$2 360.34
(C) \$2 500.00
(D) \$2 600.00

15. If \$7 000 is invested at the rate of 5% per annum compound interest for 2 years, then the TOTAL amount accruing at the end of the 2 years is

- (A) \$ 717.50
(B) \$7 700.00
(C) \$7 717.50
(D) \$7 800.00

14. If sales tax is charged at a rate of 1.5% on the marked price of an item, then the sales tax on a dress which costs \$1 800 is

- (A) \$ 27
(B) \$ 270
(C) \$ 1 827
(D) \$ 2 070

16. An article was bought for \$360 after a discount of 10% was given. The cost price of the article was

- (A) \$324
(B) \$392
(C) \$396
(D) \$400

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17. The cash price of a television set is \$350. When bought on hire purchase, a deposit of \$35 is required, followed by 12 monthly payments of \$30 each. How much is saved by paying cash?

(A) \$10
(B) \$25
(C) \$40
(D) \$45

18. If the simple interest on \$900 at 4% per annum for t years is \$108, what is the value of t ?

(A) 2
(B) 3
(C) 4
(D) 5

19. At the end of any year, a computer is worth 4% less than what it was worth at the beginning of the year. If the computer is worth \$4 200 in December 2003, then its value in January 2003 was

(A) \$4 050
(B) \$4 032
(C) \$4 368
(D) \$4 375

20. A man's basic pay for a 40-hour week is \$480. Overtime is paid at 25% in excess of the basic rate. In a certain week he works overtime and his TOTAL wage is \$660. Altogether, the number of hours he works for the week is

(A) 45
(B) 47
(C) 48
(D) 50

21. The sum of seven times the number a and the number b may be written as

(A) $7ab$
(B) $49ab$
(C) $7a + b$
(D) $7(a + b)$

22. The algebraic fraction $\frac{14a^2b^2}{45a^2b}$ written in its simplest form is

(A) $\frac{b}{3a}$
(B) $\frac{3a}{b}$
(C) $3a^2b$
(D) $\frac{1}{3} a^2b^2$

23. The value of $3^{2x+1} \times \frac{1}{9^x}$ when $x = \frac{1}{2}$ is

(A) 3
(B) 9
(C) 12
(D) 27

24. If $5(2x - 1) = 35$, then $x =$

(A) -4
(B) $\frac{1}{4}$
(C) 3
(D) 4

25. The sum of two numbers is 140. If one number is 10 less than two times the other, then the two numbers are

(A) 50 and 90
(B) 60 and 80
(C) 100 and 40
(D) 110 and 30

26. Given that $2 * 3 = 1$ and $4 * 7 = 9$, then $a * b$ may be defined as

(A) $6a - b$
(B) $a^2 - b$
(C) $2a + b$
(D) $4(b - a)$

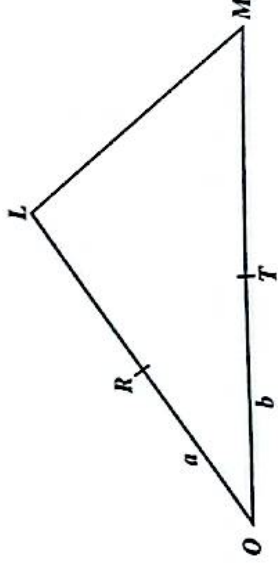
27. Given that $A = \begin{bmatrix} 1 & 3 & -3 \\ 3 & 0 & 5 \end{bmatrix}$, then $3A$ equals

(A) $\begin{bmatrix} 3 & 9 & -9 \\ 9 & 0 & 15 \end{bmatrix}$
(B) $\begin{bmatrix} 4 & 6 & -6 \\ 6 & 3 & 8 \end{bmatrix}$
(C) $\begin{bmatrix} 3 & 9 & -6 \\ 9 & 0 & 15 \end{bmatrix}$
(D) $\begin{bmatrix} -2 & 0 & 6 \\ 0 & 3 & 2 \end{bmatrix}$

28. The determinant of the 2×2 identity matrix is

(A) one
(B) zero
(C) undefined
(D) negative one

Item 29 refers to the following diagram of triangle OLM , in which R is the midpoint of OL and T the midpoint of OM . Further, $\vec{OR} = \mathbf{a}$ and $\vec{OT} = \mathbf{b}$.



29. $\vec{TL}$, expressed in terms of $\mathbf{a}$ and $\mathbf{b}$, is

(A) $\mathbf{b} + 2\mathbf{a}$
(B) $2\mathbf{a} - \mathbf{b}$
(C) $2(\mathbf{b} - \mathbf{a})$
(D) $2(\mathbf{a} + \mathbf{b})$

Item 30 refers to the following matrix, A , in which a and b are constants.

$$A = \begin{bmatrix} 2 & a \\ b & -2 \end{bmatrix}$$

30. The matrix A maps the point $P(2, 5)$ onto the point $Q(-1, 2)$. The values of a and b are

	a	b
(A)	1	6
(B)	-6	1
(C)	-1	6
(D)	-1	-6

31. Given that

$$1 \text{ centimetre} = \frac{1}{100\,000} \text{ kilometres,}$$

then 2.5 kilometres, in centimetres, is

- (A) 250
- (B) 2 500
- (C) 25 000
- (D) 250 000

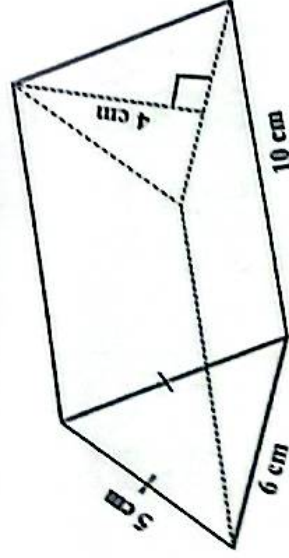
34. If the radius of a circular pond, in terms of

π , is $\frac{44}{\pi}$ metres, then the distance around

the edge of the circular pond, in metres, is

- (A) 88
- (B) 88π
- (C) 196
- (D) $1\,936\pi$

Item 32 refers to the following diagram which shows a triangular prism.



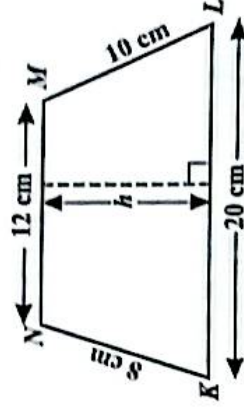
32. The surface area of the triangular prism, in cm^2 , is

- (A) 60
- (B) 124
- (C) 184
- (D) 320

35. The perimeter of a square is 56 cm. What is its area, in cm^2 ?

- (A) 28
- (B) 78
- (C) 169
- (D) 196

Item 36 refers to the following diagram of a trapezium, $KL.MN$.



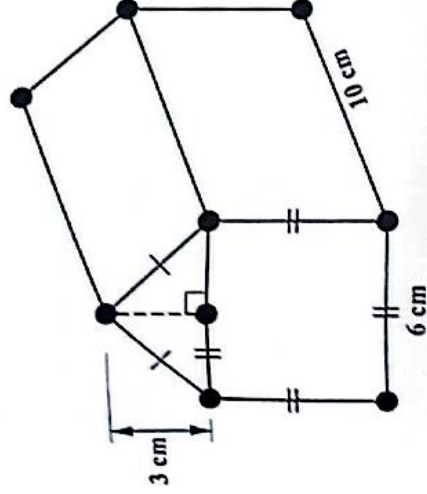
33. At a party, a number of guests were served 15 litres of champagne. Each guest had 2 glasses of champagne and each glass held 150 millilitres. Assuming no spillage, how many guests were at the party?

- (A) 10
- (B) 75
- (C) 50
- (D) 100

36. If the area of the trapezium, $KL.MN$, is 80 cm^2 , then the height, h , is

- (A) 5.0 cm
- (B) 7.3 cm
- (C) 8.0 cm
- (D) 8.5 cm

Item 37 refers to the following diagram which shows a compound solid shape that consists of a triangular prism with a height of 3 cm, resting on a rectangular-based prism.



37. The volume of the compound solid shape, in cm^3 , is

- (A) 90
- (B) 360
- (C) 450
- (D) 900

38.

When a pilot left Country A to travel to Country B, the time on his watch was 22:50 hours. The distance between Country A and B was 3 000 km, which the pilot covered at an average speed of 625 km/h for the entire journey.

Assuming that both countries are in the same time zone, what was the time in Country B when the pilot arrived the next day?

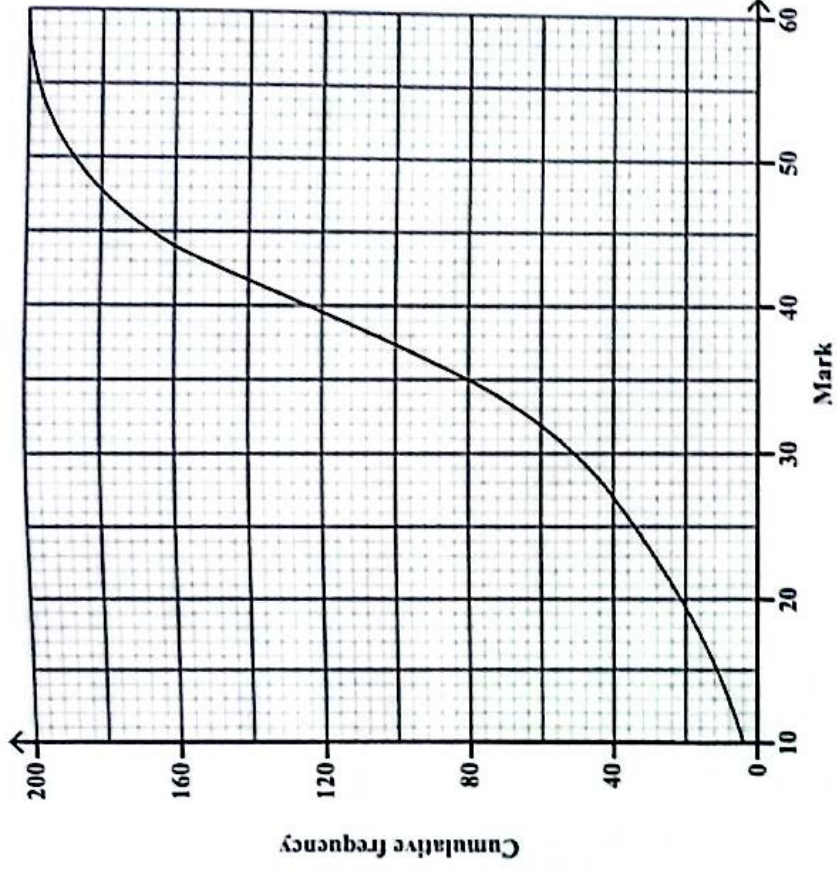
- (A) 02:50 hours
- (B) 03:38 hours
- (C) 03:50 hours
- (D) 15:38 hours

39.

Ms Ray recorded the 15 test scores of her students and, for reporting purposes, selected the score that appeared most often. Which of the following statistical measures did Ms Ray obtain?

- (A) Mean
- (B) Mode
- (C) Range
- (D) Median

Items 40 and 41 refer to the following diagram which shows the cumulative frequency curve based on the marks of 200 students who took a driving test.



40. How many students scored AT MOST 45 marks?

- (A) 34
- (B) 40
- (C) 166
- (D) 170

41. The median mark scored by the 200 students is

- (A) 30.0
- (B) 37.5
- (C) 48.5
- (D) 100.0

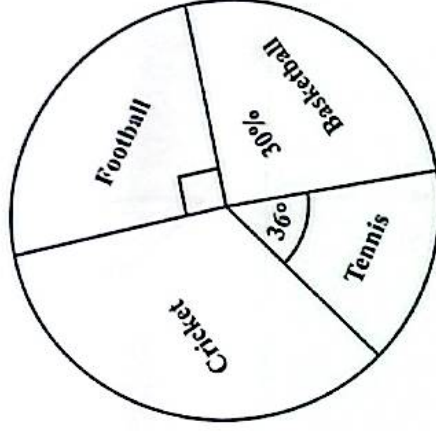
42. If the mean of the four numbers 4, 8, x and 12 is 10, then x is

(A) 4
(B) 10
(C) 12
(D) 16

45. The equation of the line that crosses the vertical axis at the point (0, 5) and that has a gradient of 4 is

(A) $y = 4x$
(B) $y = 5x$
(C) $y = 4x + 5$
(D) $y = 5x + 4$

Items 43 and 44 refer to the following pie chart which shows the popular games played by 720 students.



46. The point where a linear function crosses the vertical axis is

(A) the x -intercept
(B) the y -intercept
(C) always positive
(D) always negative

43. How many students play cricket OR football?

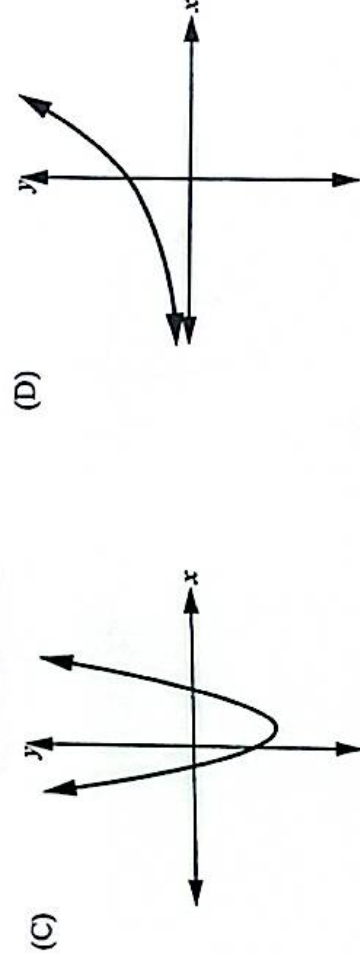
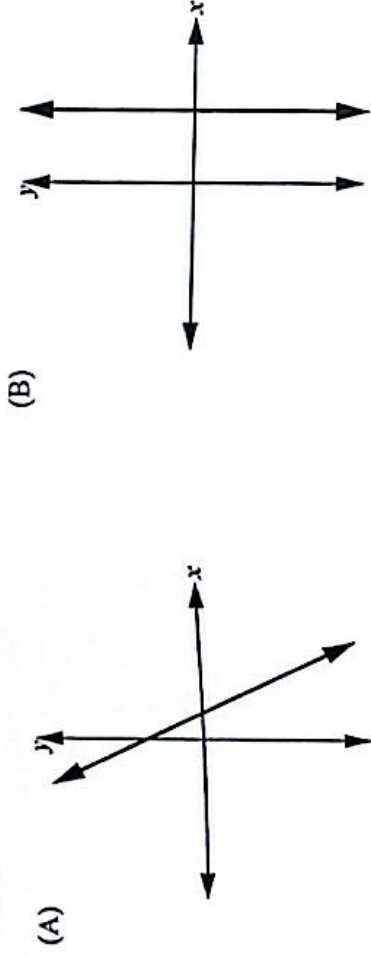
(A) 60
(B) 180
(C) 432
(D) 504

44. A student is picked at random from the group. What is the probability that the student plays NEITHER tennis NOR football?

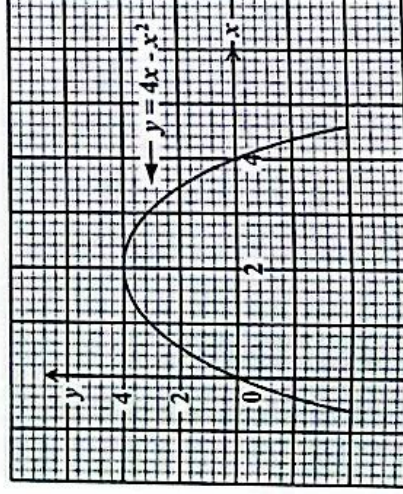
(A) $\frac{1}{5}$
(B) $\frac{3}{10}$
(C) $\frac{2}{5}$
(D) $\frac{13}{20}$

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47. Which of the following graphs represents a linear function?



Item 48 refers to the following graph of a function.



48. The coordinates of the maximum point of the function $y = 4x - x^2$ are

- (A) (0, 4)
- (B) (2, 4)
- (C) (4, 2)
- (D) (4, 4)

49. Which of the following functions is undefined at the point $x = -2$?

(A) $f(x) = \frac{3x}{2-x}$

(B) $f(x) = \frac{1}{x-2}$

(C) $f(x) = \frac{x+2}{x^2-2}$

(D) $f(x) = \frac{5}{x+2}$

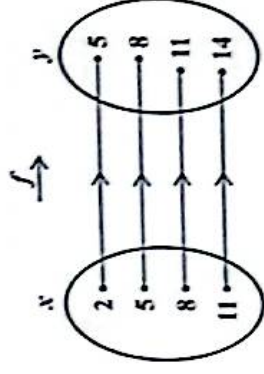
50. If $g(x) = x^2 + 1$, then $gg(-1) =$

- (A) -5
(B) 1
(C) 2
(D) 5

51. A parking lot charges \$1.00 plus \$0.25 for every hour (h) or part thereof for which a vehicle is parked. A function, $C(h)$, that represents the parking cost is

- (A) $C(h) = 0.25h$
(B) $C(h) = 0.25h + 1.00$
(C) $C(h) = 1.00h + 0.25$
(D) $C(h) = \frac{h}{0.25} + 1.00$

- Item 52 refers to the following arrow diagram which shows a function, f .



52. Which of the following equations BEST describes the function?

- (A) $y = x + 3$
(B) $x + y = 3$
(C) $x = y + 3$
(D) $y = 2x + 1$

- Item 53 refers to the following diagram of a transformation.



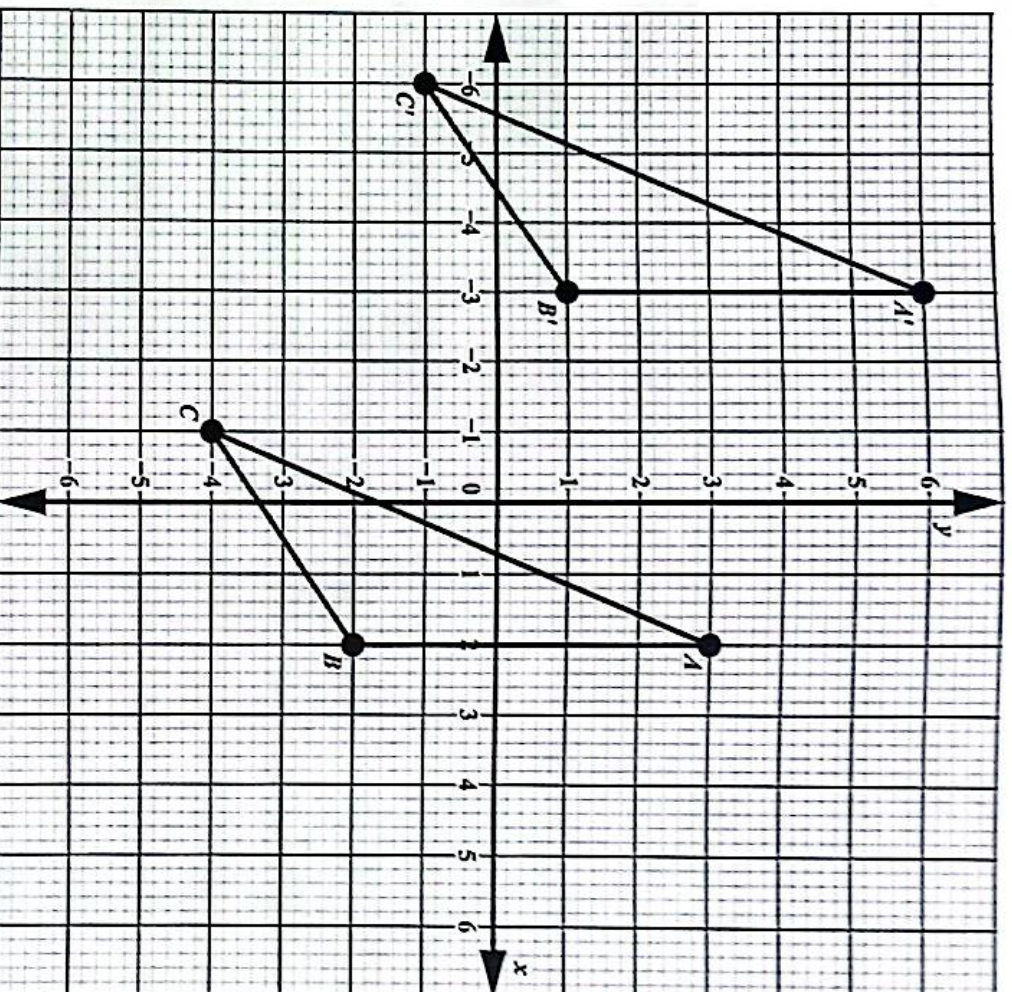
53. What transformation maps PQR onto $P'Q'R'$?

- (A) Rotation
(B) Reflection
(C) Translation
(D) Enlargement

54. In which of the following polygons is the sum of the interior angles TWICE the sum of the exterior angles?

- (A) Triangle
(B) Hexagon
(C) Pentagon
(D) Quadrilateral

Item 55 refers to the following diagram which shows the translation of Triangle ABC to Triangle $A'B'C'$.



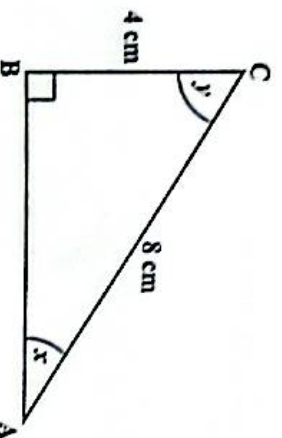
55. In the diagram, the translation by which ABC is mapped onto $A'B'C'$ is represented by

- (A) $\begin{bmatrix} -5 \\ 3 \end{bmatrix}$
- (B) $\begin{bmatrix} -5 \\ -3 \end{bmatrix}$
- (C) $\begin{bmatrix} 5 \\ -3 \end{bmatrix}$
- (D) $\begin{bmatrix} 5 \\ 3 \end{bmatrix}$

56. The point $P(2, -3)$ is rotated about the origin through an angle of 90° in an anti-clockwise direction. What are the coordinates of the image P ?

(A) $(3, 2)$
 (B) $(2, 3)$
 (C) $(-3, 2)$
 (D) $(3, -2)$

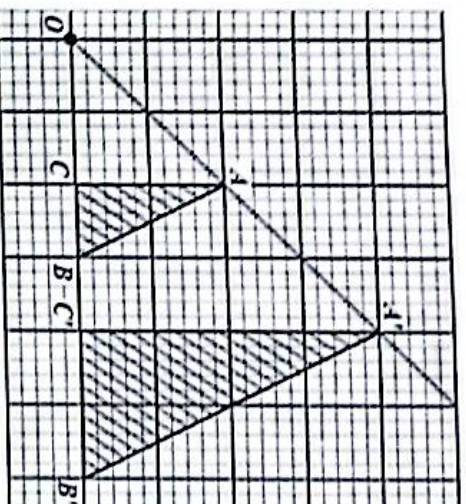
Item 57 refers to the following diagram of a right-angled triangle, $\triangle ABC$.



57. In the right-angled triangle above, which trigonometric ratio is equal to $\frac{4}{8}$?

(A) $\cos x$
 (B) $\tan x$
 (C) $\tan y'$
 (D) $\sin x$

Item 58 refers to the following diagram of an enlargement.



58. OA' , OB' and OC' are straight lines. $\triangle ABC$ is mapped onto $\triangle A'B'C'$ by an enlargement whose centre is O . What is the scale factor of the enlargement?

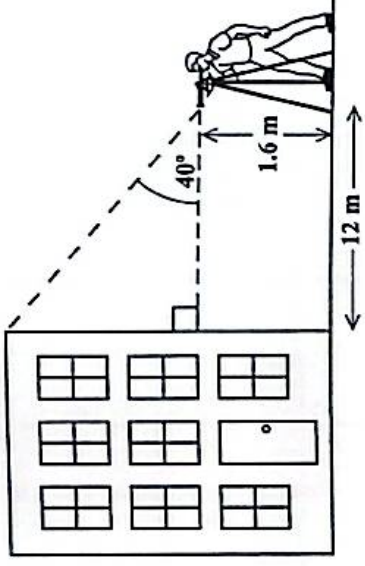
(A) $\frac{1}{2}$
 (B) $-\frac{1}{2}$
 (C) 2
 (D) -2

59. A plane is flying in the direction of 045° and changes course in a clockwise direction to 135° . The angle through which the plane turns is

(A) 45°
 (B) 90°
 (C) 135°
 (D) 270°

Item 60 refers to the following diagram of a building.

A surveyor sets up his instruments 12 metres from the foot of the building and records the angle of elevation of the top of the building.



60. An estimate of the height of the building can be obtained by calculating

- (A) $1.6 + 12 \cos 40^\circ$
- (B) $1.6 + 12 \tan 40^\circ$
- (C) $(12 \tan 40^\circ) - 1.6$
- (D) $(12 \sin 40^\circ) - 1.6$

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.